```
# ============================================
# 1. CONFIGURACIÓN DEL ENTORNO (GOOGLE COLAB)
# ============================================

!pip install -q streamlit pyngrok pandas numpy scikit-learn plotly
!pip install -q streamlit-plotly-events
# Permite ejecutar Streamlit dentro de Colab
from pyngrok import ngrok
```

```
# ============================================
# 2. IMPORTACIÓN DE LIBRERÍAS
# ============================================

import pandas as pd
import numpy as np
import streamlit as st
import plotly.express as px

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
```

```
# ============================================
# 3. CARGA DINÁMICA DEL DATASET
# ============================================

def load_dataset(uploaded_file):
    """
    Carga el dataset de forma dinámica.
    Permite reutilizar el sistema con diferentes fuentes de datos.
    """
    if uploaded_file.name.endswith('.csv'):
        return pd.read_csv(uploaded_file)
    elif uploaded_file.name.endswith('.xlsx'):
        return pd.read_excel(uploaded_file)
    else:
        st.error("Formato no soportado")
        return None
```

```
# ============================================
# 4. CÁLCULO DE VARIABLES RFM
# ============================================

def calculate_rfm(df, customer_col, date_col, amount_col):
    df[date_col] = pd.to_datetime(df[date_col])

    # Limpieza estricta: Forzar numérico para evitar colapso de sum() en variables datetime/string
    df[amount_col] = pd.to_numeric(df[amount_col], errors='coerce')
    df = df.dropna(subset=[amount_col])

    reference_date = df[date_col].max() + pd.Timedelta(days=1)

    rfm = df.groupby(customer_col).agg({
        date_col: lambda x: (reference_date - x.max()).days,
        customer_col: 'count',
        amount_col: 'sum'
    })

    rfm.columns = ['Recency', 'Frequency', 'Monetary']
    return rfm.reset_index()
```

```
# ============================================
# 5. NORMALIZACIÓN Y MÉTRICAS DE VALIDACIÓN
# ============================================

def scale_features(rfm_df):
    scaler = StandardScaler()
    scaled = scaler.fit_transform(rfm_df[['Recency','Frequency','Monetary']])
    return scaled

def calculate_elbow_silhouette(data, max_k=10):
    inertia = []
    silhouette = []
    for k in range(2, max_k+1):
        model = KMeans(n_clusters=k, random_state=42)
        labels = model.fit_predict(data)
        inertia.append(model.inertia_)
        silhouette.append(silhouette_score(data, labels))
```

```
    return inertia, silhouette
```

```
# ============================================
# 6. ENTRENAMIENTO DEL MODELO K-MEANS (K)
# ============================================

def train_kmeans(data, k):
    '''Entrena el modelo K-Means usando el K óptimo descubierto.'''
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(data)
    return kmeans, labels
```

```
# ============================================
# 7. ASIGNACIÓN DE SEGMENTOS RFM (AJUSTADA)
# ============================================

def assign_dynamic_segments(rfm):
    '''
    Mapeo dinámico: Evalúa matemáticamente los centroides descubiertos.
    Score = Monetary_z + Frequency_z - Recency_z.
    '''
    centroids = rfm.groupby('Cluster')[['Recency', 'Frequency', 'Monetary']].mean()

    scaler = StandardScaler()
    scaled_centroids = pd.DataFrame(
        scaler.fit_transform(centroids),
        columns=['R_z', 'F_z', 'M_z'],
        index=centroids.index
    )

    scaled_centroids['Score'] = scaled_centroids['M_z'] + scaled_centroids['F_z'] - scaled_centroids['R_z']
    scaled_centroids = scaled_centroids.sort_values('Score', ascending=False)

    cluster_to_label = {}
    for idx, (cluster_id, row) in enumerate(scaled_centroids.iterrows()):
        if idx == 0:
            cluster_to_label[cluster_id] = 'VIP'
        elif idx == len(scaled_centroids) - 1:
            cluster_to_label[cluster_id] = 'Perdidos'
        else:
            if row['R_z'] > 0:
                cluster_to_label[cluster_id] = 'En Riesgo'
            else:
                cluster_to_label[cluster_id] = 'Leales'

    rfm['Segmento'] = rfm['Cluster'].map(cluster_to_label)
    return rfm
```

```
# ============================================
# 8. RECOMENDACIONES ESTRATÉGICAS DE MARKETING
# ============================================

def marketing_recommendations(segment):
    recommendations = {
        "VIP": [
            "Ofrecer programas de fidelización exclusivos",
            "Acceso anticipado a promociones y lanzamientos",
            "Comunicación personalizada 1 a 1",
            "Incentivos por referidos"
        ],
        "Leales": [
            "Descuentos progresivos por frecuencia de compra",
            "Campañas de cross-selling y up-selling",
            "Programas de puntos",
            "Encuestas de satisfacción"
        ],
        "En Riesgo": [
            "Campañas de reactivación con descuentos agresivos",
            "Recordatorios personalizados por email/SMS",
            "Ofertas por tiempo limitado",
            "Análisis de causas de abandono"
        ],
        "Perdidos": [
            "Campañas de recuperación de bajo costo",
            "Ofertas de bienvenida para reenganche",
            "Excluir de campañas costosas",
            "Evaluar eliminación del CRM activo"
        ]
    }
```

```
    return recommendations.get(segment, [])
```

```
# ==========================================
# 9. DASHBOARD INTERACTIVO STREAMLIT
# ==========================================

st.title("Segmentación de Clientes – Retail (RFM + K-Means)")

uploaded_file = st.file_uploader("Cargar dataset transaccional")

if uploaded_file:
    df = load_dataset(uploaded_file)

    # Autodetección de índices para sugerir las columnas correctas
    def_cust_idx = 0
    def_date_idx = next((i for i, c in enumerate(df.columns) if 'date' in c.lower() or 'fecha' in c.lower()), 0)
    def_amt_idx = next((i for i, c in enumerate(df.columns) if 'monto' in c.lower() or 'total' in c.lower() or 'amount' in c.lower()), 0)

    col1, col2, col3 = st.columns(3)
    with col1: customer_col = st.selectbox("Columna ID Cliente", df.columns, index=def_cust_idx)
    with col2: date_col = st.selectbox("Columna Fecha", df.columns, index=def_date_idx)
    with col3: amount_col = st.selectbox("Columna Monto", df.columns, index=def_amt_idx)

    # Manejo de estado para evitar que la app se reinicie al usar el selectbox final
    if 'ejecutado' not in st.session_state:
        st.session_state.ejecutado = False

    if st.button("Ejecutar Segmentación"):
        st.session_state.ejecutado = True

    if st.session_state.ejecutado:
        try:
            rfm = calculate_rfm(df, customer_col, date_col, amount_col)
            scaled = scale_features(rfm)

            st.markdown("---")
            st.subheader("Optimización del Número de Clusters ($K$)")

            inertia, silhouette = calculate_elbow_silhouette(scaled, max_k=10)
            best_k = int(np.argmax(silhouette) + 2)

            sil_df = pd.DataFrame({'K': range(2, 11), 'Silhouette Score': silhouette})

            st.line_chart(sil_df.set_index('K'))
            st.success(f"**$K$ Óptimo matemático calculado y aplicado:** {best_k}")

            # Entrenar modelo con K dinámico automático
            model, labels = train_kmeans(scaled, best_k)
            rfm['Cluster'] = labels

            # Asignar etiquetas dinámicamente
            rfm = assign_dynamic_segments(rfm)

            st.markdown("---")
            col_bar, col_3d = st.columns([1, 2])

            with col_bar:
                st.subheader("Distribución de Segmentos")
                st.bar_chart(rfm['Segmento'].value_counts())

            with col_3d:
                st.subheader("Topología 3D")
                fig = px.scatter_3d(
                    rfm, x='Recency', y='Frequency', z='Monetary',
                    color='Segmento', opacity=0.8,
                    title="Distribución RFM por Segmento"
                )
                fig.update_layout(margin=dict(l=0, r=0, b=0, t=30))
                st.plotly_chart(fig, use_container_width=True)

            st.markdown("---")
            st.subheader("Recomendaciones de Marketing")
            segmentos_disponibles = rfm['Segmento'].unique().tolist()
            selected_segment = st.selectbox("Selecciona un segmento descubierto", segmentos_disponibles)

            recs = marketing_recommendations(selected_segment)
            for r in recs:
                st.markdown(f"✅ {r}")

        except Exception as e:
            st.error(f"🚨 Error al procesar: Verifica que seleccionaste columnas válidas para Fecha y Monto. Detalle: {e}")
            st.session_state.ejecutado = False
```

```
2026-02-23 19:53:59.512 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.513 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.514 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.515 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.517 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.519 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.520 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.522 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-23 19:53:59.522 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
```

```
# Ejecutar Streamlit en Colab
# Obtén tu authtoken de ngrok en https://dashboard.ngrok.com/get-started/your-authtoken
# y reemplaza 'YOUR_AUTHTOKEN' con tu token real.
ngrok.set_auth_token('38CulDXlM53o6OGkHScEziK918U_4TJ3Erf2M63j3FoxhJqPX')
public_url = ngrok.connect(8501)
print(public_url)

# Guardar el código de Streamlit en un archivo app.py
with open('app.py', 'w') as f:
    f.write("""
import pandas as pd
import numpy as np
import streamlit as st
import plotly.express as px

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

# ==========================================
# 3. CARGA DINÁMICA DEL DATASET
# ==========================================
def load_dataset(uploaded_file):
    \"\"\"
    Carga el dataset de forma dinámica.
    Permite reutilizar el sistema con diferentes fuentes de datos.
    \"\"\"
    if uploaded_file.name.endswith('.csv'):
        return pd.read_csv(uploaded_file)
    elif uploaded_file.name.endswith('.xlsx'):
        return pd.read_excel(uploaded_file)
    else:
        st.error("Formato no soportado")
        return None

# ==========================================
# 4. CÁLCULO DE VARIABLES RFM
# ==========================================
def calculate_rfm(df, customer_col, date_col, amount_col):
    df[date_col] = pd.to_datetime(df[date_col])

    # Limpieza estricta: Forzar numérico para evitar colapso de sum() en variables datetime/string
    df[amount_col] = pd.to_numeric(df[amount_col], errors='coerce')
    df = df.dropna(subset=[amount_col])

    reference_date = df[date_col].max() + pd.Timedelta(days=1)

    rfm = df.groupby(customer_col).agg({
        date_col: lambda x: (reference_date - x.max()).days,
        customer_col: 'count',
        amount_col: 'sum'
    })

    rfm.columns = ['Recency', 'Frequency', 'Monetary']
    return rfm.reset_index()

# ==========================================
# 5. NORMALIZACIÓN Y MÉTRICAS DE VALIDACIÓN
# ==========================================
def scale_features(rfm_df):
    scaler = StandardScaler()
    scaled = scaler.fit_transform(rfm_df[['Recency','Frequency','Monetary']])
    return scaled

def calculate_elbow_silhouette(data, max_k=10):
    inertia = []
    silhouette = []
    for k in range(2, max_k+1):
        model = KMeans(n_clusters=k, random_state=42)
        labels = model.fit_predict(data)
        inertia.append(model.inertia_)
        silhouette.append(silhouette_score(data, labels))
    return inertia, silhouette

# ==========================================
# 6. ENTRENAMIENTO DEL MODELO K-MEANS
# ==========================================
```

```
def train_kmeans(data, k):
    \"\"\"Entrena el modelo K-Means usando el K óptimo descubierto.\"\"\"
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(data)
    return kmeans, labels

# ==========================================
# 7. ASIGNACIÓN DINÁMICA DE SEGMENTOS RFM
# ==========================================
def assign_dynamic_segments(rfm):
    \"\"\"
    Mapeo dinámico: Evalúa matemáticamente los centroides descubiertos.
    Score = Monetary_z + Frequency_z - Recency_z.
    \"\"\"
    centroids = rfm.groupby('Cluster')[['Recency', 'Frequency', 'Monetary']].mean()

    scaler = StandardScaler()
    scaled_centroids = pd.DataFrame(
        scaler.fit_transform(centroids),
        columns=['R_z', 'F_z', 'M_z'],
        index=centroids.index
    )

    scaled_centroids['Score'] = scaled_centroids['M_z'] + scaled_centroids['F_z'] - scaled_centroids['R_z']
    scaled_centroids = scaled_centroids.sort_values('Score', ascending=False)

    cluster_to_label = {}
    for idx, (cluster_id, row) in enumerate(scaled_centroids.iterrows()):
        if idx == 0:
            cluster_to_label[cluster_id] = 'VIP'
        elif idx == len(scaled_centroids) - 1:
            cluster_to_label[cluster_id] = 'Perdidos'
        else:
            if row['R_z'] > 0:
                cluster_to_label[cluster_id] = 'En Riesgo'
            else:
                cluster_to_label[cluster_id] = 'Leales'

    rfm['Segmento'] = rfm['Cluster'].map(cluster_to_label)
    return rfm

# ==========================================
# 8. RECOMENDACIONES ESTRATÉGICAS DE MARKETING
# ==========================================
def marketing_recommendations(segment):
    recommendations = {
        "VIP": [
            "Programas de fidelización premium",
            "Ofertas exclusivas y lanzamientos anticipados",
            "Beneficios por referidos",
            "Atención personalizada"
        ],
        "Leales": [
            "Promociones por frecuencia",
            "Cross-selling y up-selling",
            "Programa de puntos",
            "Comunicación periódica personalizada"
        ],
        "En Riesgo": [
            "Campañas de reactivación",
            "Descuentos por tiempo limitado",
            "Recordatorios personalizados",
            "Análisis de churn"
        ],
        "Perdidos": [
            "Campañas de recuperación de bajo costo",
            "Ofertas de reenganche",
            "Excluir de campañas costosas",
            "Depuración del CRM"
        ]
    }
    return recommendations.get(segment, [])

# ==========================================
# 9. DASHBOARD INTERACTIVO STREAMLIT
# ==========================================
st.title("Segmentación de Clientes – Retail (RFM + K-Means)")

uploaded_file = st.file_uploader("Cargar dataset transaccional")

if uploaded_file:
    df = load_dataset(uploaded_file)

    # Autodetección de índices para sugerir las columnas correctas
    def_cust_idx = 0
    def_date_idx = next((i for i, c in enumerate(df.columns) if 'date' in c.lower() or 'fecha' in c.lower()), 0)
    def_amt_idx = next((i for i, c in enumerate(df.columns) if 'monto' in c.lower() or 'total' in c.lower() or 'amount' in c.lower()), 0)
```

```
            col1, col2, col3 = st.columns(3)
            with col1: customer_col = st.selectbox("Columna ID Cliente", df.columns, index=def_cust_idx)
            with col2: date_col = st.selectbox("Columna Fecha", df.columns, index=def_date_idx)
            with col3: amount_col = st.selectbox("Columna Monto", df.columns, index=def_amt_idx)

            # Manejo de estado para evitar que la app se reinicie al usar el selectbox final
            if 'ejecutado' not in st.session_state:
                st.session_state.ejecutado = False

            if st.button("Ejecutar Segmentación"):
                st.session_state.ejecutado = True

            if st.session_state.ejecutado:
                try:
                    rfm = calculate_rfm(df, customer_col, date_col, amount_col)
                    scaled = scale_features(rfm)

                    st.markdown("---")
                    st.subheader("Optimización del Número de Clusters ($K$)")

                    inertia, silhouette = calculate_elbow_silhouette(scaled, max_k=10)
                    best_k = int(np.argmax(silhouette) + 2)

                    sil_df = pd.DataFrame({'K': range(2, 11), 'Silhouette Score': silhouette})

                    st.line_chart(sil_df.set_index('K'))
                    st.success(f"**$K$ Óptimo matemático calculado y aplicado:** {best_k}")

                    # Entrenar modelo con K dinámico automático
                    model, labels = train_kmeans(scaled, best_k)
                    rfm['Cluster'] = labels

                    # Asignar etiquetas dinámicamente
                    rfm = assign_dynamic_segments(rfm)

                    st.markdown("---")
                    col_bar, col_3d = st.columns([1, 2])

                    with col_bar:
                        st.subheader("Distribución de Segmentos")
                        st.bar_chart(rfm['Segmento'].value_counts())

                    with col_3d:
                        st.subheader("Topología 3D")
                        fig = px.scatter_3d(
                            rfm, x='Recency', y='Frequency', z='Monetary',
                            color='Segmento', opacity=0.8,
                            title="Distribución RFM por Segmento"
                        )
                        fig.update_layout(margin=dict(l=0, r=0, b=0, t=30))
                        st.plotly_chart(fig, use_container_width=True)

                    st.markdown("---")
                    st.subheader("Recomendaciones de Marketing")
                    segmentos_disponibles = rfm['Segmento'].unique().tolist()
                    selected_segment = st.selectbox("Selecciona un segmento descubierto", segmentos_disponibles)

                    recs = marketing_recommendations(selected_segment)
                    for r in recs:
                        st.markdown(f"✅ {r}")

                except Exception as e:
                    st.error(f"🚨 Error al procesar: Verifica que seleccionaste columnas válidas para Fecha y Monto. Detalle: {e}")
                    st.session_state.ejecutado = False
""")

# Ahora, ejecuta el archivo app.py con streamlit
!streamlit run app.py
```

```
NgrokTunnel: "https://overcoolly-cycadaceous-colten.ngrok-free.dev" -> "http://localhost:8501"

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.


  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://172.28.0.12:8501
  External URL: http://35.237.13.132:8501

2026-02-23 19:59:06.364 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
  Stopping...
  Stopping...
```

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.